

On considère la suite  $\{u_n = 2n + 1\}$  définie sur  $n \in \mathbb{N}$ . On veut calculer la somme des  $u_n$  :

*Démonstration.*

$$\begin{aligned}\sum_{k=0}^n u_k &= \frac{(u_0 + u_n)(n+1)}{2} \\ &= \frac{(2n+2)(n+1)}{2} \\ &= \frac{2(n+1)(n+1)}{2} \\ &= (n+1)(n+1) \\ &= (n+1)^2\end{aligned}$$

□

$$\begin{aligned}pH &= -\log\left(\frac{[H_3O_{(aq)}^+]}{c_0}\right) \\ [H_3O_{(aq)}^+] &= 10^{-pH} * c_0\end{aligned}$$

Euler's formula :  $e^{\theta i} = \cos(\theta) + i \sin(\theta)$

$$\begin{aligned}x^2 - 2xy &= 15 \\ x^2 &= 15 + 2xy\end{aligned}$$

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

Example :

$$A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Exercice 2)

$$\begin{aligned}\sqrt{2}y' &= \sqrt{6}y - 1 \\ y &= Ce^{\frac{\sqrt{12}}{2}x} + \frac{1}{\sqrt{6}}\end{aligned}$$